

Communication-avoiding CG variants for the FESOM2 SSH solve

Derivation cross-check and discrepancy report

Date	6 August 2026
Scope	cg2, pipecg, oati, pcsi — re-derived, then cross-checked against four sources
Purpose	Report the discrepancies found, and one that explains an observed instability
Detail	docs/plans/20260805-m10-ssh-derivations.md, branch m10-ssh-solvers

1. What this is, and the one result that matters

Four communication-avoiding solvers for the FESOM2 sea-surface-height system were implemented for the Kokkos port. Rather than transcribe the published listings, each was re-derived from first principles and then compared against the papers and against the existing Fortran implementation. Nine discrepancies came out of that comparison; they are tabulated in §3.

One of them is not a typo but a missing hypothesis, and it matters enough to lead with:

The α recurrence used by every Chronopoulos–Gear variant is valid only when the preconditioner is symmetric. None of the sources say so. The FESOM MITgcm-class preconditioner is not symmetric, and the resulting error in the step length reaches 21.8% on the production CORE2 matrix.

Running cg2 with the preconditioner as built makes the solver diverge on 18 of 20 solves. Symmetrising it — a one-line change that provably preserves the preconditioner’s spectrum — removes the divergence entirely. This reproduces and explains the cg2 instability that had been observed previously. §2 gives the derivation, the measurements and the remedy.

Sources referred to

- [S] S. Danilov, solvers.F90 — ssh_solve_cg2, ssh_solve_cg_pipelined.
- [CV] Cools & Vanroose, arXiv:1612.01395v3 — Alg. 4 (Chronopoulos–Gear CG), Alg. 6 (pipelined CG).
- [P] Huang *et al.*, *Geosci. Model Dev.* **9**, 4209–4225 (2016) — App. B2, B3, C.
- [T] Tiwari & Vadhiyar, *J. Parallel Distrib. Comput.* **163** (2022) 147–155 — Alg. 3–5.
- [GV] Golub & Varga (1961); Saad, *Iterative Methods*, Alg. 12.1.

Notation: A is the SSH stiffness matrix (symmetric to 1.4×10^{-13} , measured), M^{-1} the applied preconditioner, r the residual, $u = M^{-1}r$, $w = Au$, $\gamma = (r, u)$, $\delta = (w, u)$.

2. The unstated symmetry hypothesis

2.1 What the code builds

fesom_ssh_preconditioner constructs, per row i ,

$$\text{pr}[i, i] = \frac{1}{d_i}, \quad \text{pr}[i, j] = -\frac{1}{2} \frac{a_{ij}/d_i}{d_i + d_j} \quad (j \neq i), \quad d_i \equiv A_{ii}. \quad (1)$$

The transpose entry carries d_j in the leading division, so $\text{pr}[i, j]/\text{pr}[j, i] = d_j/d_i$ and $M^{-1} \neq (M^{-1})^\top$ wherever $d_i \neq d_j$. On an unstructured ocean mesh the diagonal varies with cell area and depth, so this is structural, not a small perturbation. Measured on CORE2:

operator	$\max X_{ij} - X_{ji} $	$\max X_{ij} $	defect ratio
M^{-1} (pr_values)	8.561×10^{-8}	1.342×10^{-7}	0.638
A (control)	2.150×10^{-5}	1.510×10^8	1.42×10^{-13}

2.2 Why the baseline PCG does not care, and the variants do

Baseline PCG forms $\alpha_i = \gamma_i / (p_i, Ap_i)$ with an explicit dot product every iteration. Nothing is recurrent, no orthogonality identity is invoked, and it converges normally. The Chronopoulos–Gear family replaces that explicit quantity with

$$\sigma_i = \delta_i - \beta_i^2 \sigma_{i-1}, \quad \alpha_i = \gamma_i / \sigma_i, \quad \sigma_i \equiv (p_i, Ap_i). \quad (2)$$

With $p_i = u_i + \beta_i p_{i-1}$ and A symmetric, $(p_i, Ap_i) = \delta_i + 2\beta_i(u_i, s_{i-1}) + \beta_i^2 \sigma_{i-1}$ where $s_{i-1} = Ap_{i-1}$, so (2) holds if and only if

$$(u_i, s_{i-1}) = -\beta_i \sigma_{i-1}. \quad (3)$$

Using $s_{i-1} = (r_{i-1} - r_i) / \alpha_{i-1}$,

$$(u_i, s_{i-1}) = \frac{(M^{-1}r_i, r_{i-1}) - \gamma_i}{\alpha_{i-1}}, \quad (M^{-1}r_i, r_{i-1}) = (r_i, M^{-\top}r_{i-1}). \quad (4)$$

That term vanishes by the standard PCG residual M -orthogonality *only when* $M^{-\top} = M^{-1}$; then $(u_i, s_{i-1}) = -\gamma_i / \alpha_{i-1} = -\beta_i \sigma_{i-1}$, which is (3). With a non-symmetric M^{-1} it does not vanish, (3) fails, and σ_i drifts from the true (p_i, Ap_i) — cumulatively, because σ is a recurrence. A wrong σ gives a wrong α , which is no longer the line-search minimiser: the method stops being CG.

2.3 What it costs in practice

A diagnostic replay carries both the true (p_i, Ap_i) and the recurrence (2) through a reference PCG solve on a dumped CORE2 system ($n_{\text{nz}} = 870\,146$), first with the preconditioner as built, then with the symmetrised form of §2.4:

iteration	as built: rel. σ drift	as built: $ (u_i, r_{i-1}) /\gamma$	symmetrised: drift
2	1.98×10^{-2}	1.28×10^{-2}	5.4×10^{-14}
8	8.20×10^{-2}	8.11×10^{-2}	8.3×10^{-14}
20	1.53×10^{-1}	1.66×10^{-1}	5.4×10^{-14}
40	1.77×10^{-1}	2.00×10^{-1}	3.8×10^{-14}
60	2.18×10^{-1}	2.41×10^{-1}	1.5×10^{-14}
worst	21.8 %	24.1 %	1.17×10^{-13}

The quantity required to vanish instead reaches 24 % of γ , and the step length is wrong by up to 22 % — at the production tolerance, on the production matrix. In the model, **cg2** with the literal preconditioner fires the auto-fallback guard on **18 of 20 solves**, the residual growing to 2.9×10^4 against a tolerance of ≈ 4 . With the symmetrised preconditioner: **no fallbacks**, 123.35 iterations per solve. Independently, the α sequence departs from reference PCG by 2.017 % at iteration 2, matching the measured σ drift of 1.977 % term for term.

2.4 Remedy

Factor the preconditioner as $M^{-1} = D^{-1}C$ with $D = \text{diag}(A)$, $C_{ii} = 1$, $C_{ij} = -\frac{1}{2}a_{ij}/(d_i + d_j)$. Since $a_{ij} = a_{ji}$, the matrix C is symmetric — the whole asymmetry sits in the left factor D^{-1} . Define

$$\widetilde{M}^{-1} := D^{-1/2} C D^{-1/2} \quad \text{i.e.} \quad \widetilde{\text{pr}}[i, j] = \text{pr}[i, j] \sqrt{d_i/d_j}, \quad \widetilde{\text{pr}}[i, i] = \text{pr}[i, i]. \quad (5)$$

$\widetilde{M}^{-1}$ is symmetric by construction, and it is *similar* to M^{-1} because $D^{1/2}(D^{-1}C)D^{-1/2} = D^{-1/2}CD^{-1/2}$. Similar matrices share their spectrum, so this is exactly the same quality of approximate inverse — symmetry is not being bought at the price of conditioning. (Verified: $\lambda_{\max}(M^{-1})$ and $\lambda_{\max}(\widetilde{M}^{-1})$ agree to ten significant figures.) Cost is one multiplication per stored entry, applied once at setup; sparsity and all halo machinery are unchanged.

One caveat worth stating: the similarity is a statement about M^{-1} , not about the preconditioned operator. $\widetilde{M}^{-1}A$ is not similar to $M^{-1}A$, and its extreme eigenvalue shifts by 0.23 % on a variable-diagonal test problem. Same convergence class, not the identical operator.

3. The discrepancies

Severity: **S1** — wrong results if transcribed literally; **S2** — misleading or not executable as written; **S3** — cosmetic.

id	source	issue	sev.
T-1	[CV] Alg. 4; [S]; [T]	The α recurrence $\sigma_i = \delta_i - \beta_i^2 \sigma_{i-1}$ is stated without its hypothesis. It is valid only for a symmetric M^{-1} (§2). The sources assume an “SPD preconditioner” in prose; the FESOM <code>pr_values</code> is not symmetric (defect ratio 0.638). Measured: α wrong by up to 21.8 % on CORE2, <code>cg2</code> diverging on 18 of 20 solves.	S1
T-2	[P] App. B2	Line 4 computes σ_k using β_k , which is only defined on line 5 — the listing is not executable in the order printed. Harmless in effect ($\sigma_0 = 0$ makes the undefined β_1 multiply zero on the first pass), but from $k \geq 2$ the reader must reorder. Similarly $\rho_0 = 1$ gives $\beta_1 = \rho_1$ rather than 0, harmless only because $s_0 = p_0 = 0$.	S2
T-3	[T] Alg. 4, l. 11	The γ_{i+1} recurrence folds (r_i, q_i) and (s_i, u_i) into one $2\alpha_i(\cdot)$ term. That identification needs $(r_i, M^{-1}w_i) = (M^{-1}r_i, w_i)$, i.e. a symmetric M^{-1} — the same unstated hypothesis as T-1.	S1
T-4	[T] Alg. 4, l. 12	$\delta_{i+1} = \lambda_8 - \alpha_i(\lambda_1 + \beta_i \lambda_2) - \alpha_i(\lambda_1 + \beta_i \lambda_2) + \dots$ — the identical product printed twice instead of $2\alpha_i(\cdot)$. Arithmetically the same, but it conceals that the two terms come from <i>different</i> inner products which coincide only under a symmetric M^{-1} .	S3
T-5	[P] App. C	The Lanczos start vector is printed as $q_1 = r_0 / (r_0^\top s_0)$: normalised by the inner product rather than by its square root . The missing root corrupts α_1 and plants a spurious small eigenvalue in T . On a 1024-unknown fixture $\theta_{\min}$ comes out as 2.49×10^{-6} instead of 5.65×10^{-3} — 2270× too small , and it does not wash out with m (same error at $m = 30$ and $m = 120$). Feeding that ν to P-CSI inflates the assumed condition number $\sim 2000\times$, so the Chebyshev polynomial is optimised for a far too wide interval and convergence is much slower than it should be — silently, since it still converges.	S1
T-6	[P] App. B3, l. 1	The ω recurrence coefficient is typeset as a fraction readable as either $\frac{1}{4}\alpha^2$ or $1/(4\alpha^2)$. <i>Not a defect in the paper</i> — a transcription hazard. Resolved to $1/(4\alpha^2)$ three ways (§4). Recorded so it need not be re-litigated.	S1 <i>if mis-read</i>
T-7	[T] §2.2	The method is designed and benchmarked with Jacobi (diagonal) preconditioning, where M^{-1} needs no halo exchange. Not a typo, but a portability caveat: with a sparse M^{-1} carrying A ’s stencil every preconditioner application consumes a halo ring, the $n \rightarrow g \rightarrow h \rightarrow e \rightarrow f$ chain becomes four chained applications, and the exchanged halo volume roughly doubles.	S2

id	source	issue	sev.
T-8	[S] both routines	MPI_Iallreduce is followed immediately by MPI_Wait (ssh_solve_cg2: 37–38, 61–62, 99–100). A non-blocking collective with a zero-length overlap window is a blocking reduction <i>plus</i> the non-blocking surcharge, measured at $+8\mu\text{s}$ on openmpi/4.1.2 and $+1.6\mu\text{s}$ on openmpi/4.1.5-nvhpc.	S3 <i>perf</i>
T-9	[S] both routines	Hard-coded Do iter=1,200 with no handling of iteration-limit exhaustion: a non-converged solve exits silently carrying whatever X holds.	S2

Nothing in [CV] Alg. 4 or Alg. 6 was found to be transcriptionally wrong; both listings reproduce the derivations line for line. The substantive items are the unstated symmetry hypothesis (T-1, T-3) and the P-CSI Lanczos normalisation (T-5).

4. Resolution of the P-CSI ω recurrence (T-6)

Let the preconditioned operator have spectrum in $[\nu, \mu]$ and set $\theta = \frac{1}{2}(\mu + \nu)$, $\delta_h = \frac{1}{2}(\mu - \nu)$, $\sigma_1 = \theta/\delta_h$. The standard three-term form [GV] is $\rho_0 = 1/\sigma_1$, $\rho_k = 1/(2\sigma_1 - \rho_{k-1})$, $\Delta x_k = \rho_k \rho_{k-1} \Delta x_{k-1} + (2\rho_k/\delta_h)M^{-1}r_k$. With the paper’s variables $\alpha = 2/(\mu - \nu) = 1/\delta_h$ and $\gamma = \theta$, put $\omega_k := 2\rho_k/\delta_h$. Then

$$\omega_k = \frac{2}{\delta_h(2\sigma_1 - \rho_{k-1})} = \frac{2}{2\theta - \delta_h\rho_{k-1}}, \quad \rho_{k-1} = \frac{1}{2}\delta_h\omega_{k-1}, \quad (6)$$

so that, using $\delta_h = 1/\alpha$,

$$\omega_k = \frac{1}{\theta - \frac{1}{4}\delta_h^2\omega_{k-1}} = \frac{1}{\gamma - \frac{\omega_{k-1}}{4\alpha^2}}. \quad (7)$$

The reading $\frac{1}{4}\alpha^2$ is therefore incorrect. Two independent confirmations: the paper’s own seed $\omega_0 = 2/\gamma$ follows from $\rho_0 = 1/\sigma_1$ only under this reading; and numerically, the derived coefficient meets the Golub–Varga convergence bound at $k = 10, 20, 40$ while the misread one does not. Note also that the initial step uses $\Delta x_0 = \gamma^{-1}M^{-1}r_0$, *not* ω_0 — a reader assuming otherwise is off by exactly a factor of two on the first step.

5. How to check any of this

Everything above is reproducible from the `m10-ssh-solvers` branch:

- symmetry defect and σ drift on real dumped matrices — `tools/fesom_ssh_lab` with `-sym-check` or `-sigma-drift`;
- algorithm equivalence, Chebyshev rate, the Lanczos normalisation (T-5) and the ω resolution (T-6) — `tests/test_ssh_solvers.cpp`, 18 assertions, host-only, runs in under a second;
- non-blocking-collective progression (T-8) — `tools/iallreduce_probe.c`, run on both production MPI stacks.

Full derivations for all four solvers, the cross-source tables and the measurement provenance are in `docs/plans/20260805-m10-ssh-derivations.md`; the gate and measurement ledger is `docs/SSH_SOLVERS_M10.md`.